

Homework 1: August 2025

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You do not have to submit this assignment.

The Binomial coefficient nC_k is also denoted $\binom{n}{k}$.

Exercise 1.1. For each of the following distributions, find the mean, variance, and moment generating function $\mathbb{E}[e^{\lambda(X-\mathbb{E}X)}]$ (for $\lambda > 0$) from first principles.

- Bernoulli(p), for $0 \leq p \leq 1$:

$$p_X(0) = 1 - p, \text{ and } p_X(1) = p$$

- Binomial(n, p) for $0 \leq p \leq 1$ and positive integer n :

$$p_X(i) = \binom{n}{i} p^i (1-p)^{n-i}$$

for $i \in \{0, 1, \dots, n\}$

- Poisson(α), for $\alpha > 0$:

$$p_X(i) = \frac{\alpha^i e^{-\alpha}}{i!}, \text{ for } i = 0, 1, 2, 3, \dots$$

- Uniform($-\alpha, \alpha$):

$$f_X(x) = \begin{cases} \frac{1}{2\alpha}, & \text{if } x \in [-\alpha, \alpha] \\ 0, & \text{else} \end{cases}$$

- Gaussian $\mathcal{N}(0, \sigma^2)$:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/(2\sigma^2)}$$

- Laplace(b), for $b > 0$:

$$f_X(x) = \frac{e^{-|x|/b}}{2b}$$

- Gamma(k, θ) for $k > 0, \theta > 0$:

$$f_X(x) = \frac{x^{k-1} e^{-x/\theta}}{\Gamma(k) \theta^k}$$

where $\Gamma(\cdot)$ is the Gamma function¹

For the exercises below, you may use Stirling's approximation without proof.

$$\sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n+1}} \leq \Gamma(n+1) \leq \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n}}$$

Recall that for a positive integer n , we have $\Gamma(n+1) = n!$.

¹https://en.wikipedia.org/wiki/Gamma_function

Exercise 1.2. Fix $0 < p < 1/2$, and $\delta > 0$ such that $p(1 + \delta) < 1/2$. For a given positive integer n , choose $k = \lfloor np \rfloor$. Prove that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log_2 \binom{n}{k} = H_2(p),$$

where $H_2(p) = -p \log_2 p - (1 - p) \log_2 (1 - p)$. Use this to show the following: if $X_1, X_2, \dots, X_n$ are i.i.d. Bernoulli(p) random variables, then

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\Pr \left[\frac{1}{n} \sum_{i=1}^n X_i \geq p(1 + \delta) \right] \right) \leq -D_2(p(1 + \delta) \| p)$$

where

$$D_2(p \| q) = p \log_2 \left(\frac{p}{q} \right) + (1 - p) \log_2 \left(\frac{1 - p}{1 - q} \right).$$

Hint: derive the exact expression for the probability above, use Stirling's approximation for all the terms, and try to bound the summation as the largest term $\times$ the number of terms.

This exercise shows that for large n , the probability that the fraction of heads in n tosses of a coin deviates from p is in fact exponentially decaying in n .

The following exercise shows that sometimes our intuition on how large certain objects are, can fail in high dimensions.

Exercise 1.3. The volume of an n -dimensional ball of radius r , defined as

$$\mathcal{B}_n(r) = \{\underline{x} \in \mathbb{R}^n : \|\underline{x}\|_2 \leq r\},$$

is equal to

$$V_n(r) = \frac{\pi^{n/2} r^n}{\Gamma(n/2 + 1)}$$

Let $C_n(a)$ denote the n -dimensional cube of side length a :

$$C_n(a) = \{\underline{x} \in \mathbb{R}^n : x_i \in [-a/2, a/2] \text{ for } i = 1, 2, \dots, n\}$$

Find the radius of the ball that is inscribed within $C_n(2)$ (i.e., the largest ball that is contained within $C_n(2)$), and the volume of this ball. Compare this with the volume of the cube. How do these behave as $n \rightarrow \infty$?

Now find the radius of the ball that circumscribes $C_n(2)$ (i.e., the smallest ball that contains $C_n(2)$) and compare this with the previous result. What is the volume of this ball? How do these behave as $n \rightarrow \infty$?

Exercise 1.4. Let $X_1, \dots, X_n$ be a sequence of iid random variables with (finite) mean μ and (finite) variance σ^2 . Define $S_n = \frac{1}{n} \sum_{i=1}^n X_i$. Prove that

$$\mathbb{E}[|S_n - \mu|] \leq \frac{c}{\sqrt{n}}$$

for all sufficiently large n , where c is a constant that could depend on μ, σ^2 .

Exercise 1.5. Consider the random experiment where you toss a fair coin infinitely many times. Consider a random variable X defined in the following manner. For a given realization $\omega \in \Omega$ of the coin tosses,

$$X(\omega) = \sum_{i=1}^{\infty} 2^{-i} 1_{\{\text{The } i\text{'th coin toss is a head}\}}$$

where $1_{\{\text{The } i\text{'th coin toss is a head}\}}$ is a random variable that takes the value 1 if the i 'th coin toss is a head, and zero otherwise. Find the cdf of X .

Exercise 1.6. Let $X_1, \dots, X_n$ be an iid sequence of Bernoulli(p) random variables. Define

$$S_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Prove that

$$\Pr[|S_n - p| \geq \delta p] \leq 2e^{-\delta^2 p/3}$$

for every $0 < \delta < 1$.

Exercise 1.7. Let $X_1, \dots, X_n$ be independent (not necessarily identically distributed) continuous non-negative random variables with probability density functions $f_{X_1}, \dots, f_{X_n}$. Assume that $f_{X_i}(x) \leq 1$ for all x, i .

- Show that

$$\mathbb{E}[e^{-tX_i}] \leq \frac{1}{t}, \quad \text{for all } t > 0$$

- Show that for every $\epsilon > 0$,

$$\Pr\left[\sum_{i=1}^n X_i \leq \epsilon n\right] \leq (e\epsilon)^n$$

Exercise 1.8. Suppose that we want to estimate the mean of a distribution from iid samples $X_1, \dots, X_n$. We want to get estimates that have an error at most epsilon, i.e., if $f(X_1, \dots, X_n)$ is one estimate, then we want $f(X_1, \dots, X_n) \in (\mu - \epsilon, \mu + \epsilon)$ with probability at least $1 - \delta$. Here, ϵ, δ are parameters that decide the quality of the estimate.

Design an estimator that achieves this using

$$n = c \frac{\sigma^2}{\epsilon^2} \log \frac{1}{\delta},$$

samples, where σ^2 is the variance of X_1 and c is a universal constant.

Exercise 1.9. Prove that if X is a nonnegative random variable,

$$\mathbb{E}[X] = \int_0^\infty \Pr[X > t] dt$$

For a general random variable X , prove that

$$\mathbb{E}[X] = \int_0^\infty \Pr[X > t] dt - \int_{-\infty}^0 \Pr[X < t] dt$$

For any $p \in (0, \infty)$, show that

$$\mathbb{E}[|X|^p] = \int_0^\infty p t^{p-1} \Pr[|X| > t] dt$$

assuming that the right hand side is finite.